

Pasquale J. Montò

Software Engineer

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[Portfolio](#)

EXPERIENCE

Clinequal - Lead Software Engineer & Co-Founder

Naples ITA | Jul 2025 - Present | clinequal.com | [LinkedIn](#)

- **System Architecture:** Architected a Python/FastAPI multi-tenant clinical-trial analysis platform, separating authentication, ingestion, statistical analysis and persistence into independently testable modules.
- **Clinical Bias Analysis & Explainable AI:** Developed a plugin-based statistical engine executing 15+ tests (e.g., Kaplan-Meier, chi-squared/Fisher, KS, standardized mean differences) and implemented nine bias analyzers; paired the engine with an XAI layer that auto-generates regulatory-aligned clinical narratives and visual evidence using representation ratios and diversity metrics.
- **Privacy-Preserving Data Ingestion:** Designed production infrastructure around PostgreSQL RLS, JWT RBAC, Redis rate limiting, S3-compatible storage and immutable audit logs; validated core analysis and ingestion workflows with 103 automated tests.
- **Technical Pitching & Stakeholder Communication:** Represented the company at international tech summits (e.g., Tech Arena Stockholm), successfully translating our complex AI architecture, SDTM pipelines, and regulatory compliance to VC funds and investment banks to drive our €400K pre-seed round.

IdeaSolutions - Software Engineer

Naples ITA | Mar 2023 - Dec 2023

- **Backend Microservices Development:** Developed backend services for SuperGuidaTV, a streaming-guide platform with 500,000+ monthly users, working across a TypeScript/Node.js microservice architecture built on AWS Lambda, DynamoDB, SQS, S3, API Gateway, and Step Functions.
- **Audit Logging & Change Tracking:** Built a property-level audit system for movies, series, programs, and episodes, detecting changes across nested objects and arrays and recording the responsible operator, timestamp, entity hierarchy, and modified fields in DynamoDB.
- **Watchlist Platform:** Implemented a shareable multi-watchlist system across the DynamoDB data-access, core-service, and API layers, supporting public/private lists, follows, imports, ranking by follower count, pagination, and asynchronous aggregation through SQS-triggered Lambda functions.
- **Recommendation System Integration:** Integrated content create, update, and delete operations with the recommendation service, selectively synchronizing changed metadata and marking deleted titles unavailable through Amazon Personalize.
- **Backend Platform & Content Services:** Delivered additional production tooling including S3-backed versioned "What's New" content APIs, image-composition functionality for personalized watchlist covers, and scraper normalization for inconsistent programme titles and descriptions.

PROJECTS

Playlist Converter

- Built a full-stack application that converts playlists bidirectionally between Spotify and Apple Music, integrating Spotify OAuth 2.0, Apple Music JWT/MusicKit authentication, Redis-backed token storage, and Server-Sent Events for live conversion progress.
- Designed an extensible provider architecture using Strategy, Adapter, Abstract Factory, and Template Method patterns; normalized incompatible API models and matched tracks through ISRC lookup with metadata-similarity fallback, pagination, throttling, and exponential-backoff retries - [GitHub Link](#)

Hive Real Time Monitoring System

- Led a cross-functional team from beekeeper interviews and academic research to a prototype for non-invasive hive monitoring, investigating acoustic and vibration patterns as indicators of colony state and external threats.
- Designed the sensing architecture around ESP32, MEMS microphones and accelerometers, using local frequency-domain signal processing and MQTT to stream processed hive telemetry to a backend and SwiftUI monitoring application. [Controller](#) | [Frontend](#)

ACHIEVEMENTS

StartCup Campania Winner - Regional Startup Competition, the win led us to the PNI national competition in Ferrara.

That's a Hackathon Winner - Hackathon promoted by the US Embassy in Naples.

Press Coverage ([il Mattino](#)) - Clinequal featured for innovation in healthcare AI and clinical trial analysis.

SKILLS

- **Languages:** TypeScript, Swift, Go, Python, C/C++, Java, SQL
- **Technologies:** Node.js, PostgreSQL, DynamoDB, Redis, Docker, AWS (Lambda, SQS, S3, Step Functions)
- **Concepts:** Microservices, REST APIs, Distributed Systems, Git, Agile

LANGUAGES

- English (Fluent)
- Portuguese (Fluent)
- Italian (Fluent)
- French (Conversational)
- Spanish (Conversational)

EDUCATION

Apple Developer Academy

Naples
Sep 2024 - Jun 2025

iOS development with Swift/SwiftUI, cross-functional collaboration using Agile methodologies focusing also on a business oriented approach.

Bachelor Of Science Computer Science

Università di Napoli Federico II
Sep 2023 - Sep 2026

ERASMUS+ Exchange

Jagiellonian University Kraków
Oct 2025 - Feb 2026

Coursework: Distributed Systems, Computer Networks, Formal Language Theory, Design Patterns